



Vinyl ethylene carbonate (VEC)

ELECTROLYTE ADDITIVE

PRODUCT OVERVIEW

Vinyl ethylene carbonate (VEC) is a cyclic carbonate electrolyte additive used in lithium-ion battery research. Applications include electrolyte formulation, additive screening, and electrode-electrolyte compatibility studies.

CAS No. 4427-96-7 | Formula $C_5H_6O_3$ | Abbreviation VEC

TECHNICAL SPECIFICATIONS

PROPERTY	SPECIFICATION	REPRESENTATIVE RESULT
Appearance	Colorless transparent liquid, no mechanical impurities, insoluble matter, or flocculent material	Conforms
Color	≤20 Hazen	8.6 Hazen
Water	≤50 ppm	41.7 ppm
Assay	≥99.9 wt%	99.962 wt%
Chloride	≤5 ppm	<5.0 ppm
Sulfate	≤10 ppm	<10.0 ppm

TRACE METALS

ELEMENT	SPEC. (ppm)	REP. RESULT (ppm)	ELEMENT	SPEC. (ppm)	REP. RESULT (ppm)
Na	≤5	0.31	Mg	≤5	0.02
Ca	≤5	0.07	Cr	≤5	0.05
K	≤5	0.39	Ni	≤5	ND
Pb	≤5	0.17	Mn	≤5	ND
Fe	≤5	0.24	Ba	≤5	ND
Cu	≤5	0.12	Cd	≤5	0.04
Zn	≤5	0.08	Al	≤5	0.12

QUALITY / DOCUMENTATION

Lot-specific Certificate of Analysis and SDS are available upon request.